A tidal screen with one constant fixed by the Oort cloud

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Abstract

A two-field modification of Newtonian gravity can be weak in the solar system and strong in galaxies if the switch is the tide, not a single extra gradient. The field equation below has one free number, a constant C in the Lagrangian. Setting that constant to zero empties the Oort cloud. The value that puts the cloud back, $C = 0.579125$ at the adopted Galactic field, is then held fixed. At that value the Cassini quadrupole does not move, the Gaia wide-binary sample prefers the model to Newton, 171 SPARC galaxies fit as they do without the extra shell term, and four dwarf galaxies need only stellar mass-to-light ratios. One scalar field cannot do this job: it has one gradient at the Sun, and the Sun needs two different external fields at once.

1 The equation

The Newtonian potential Φ_N is kept. A second potential obeys

$$Q_s = z + S(x)\big(Q(z) - z - C\big), \tag{1}$$

with $z = |\nabla\Phi_N|/a_0$ and

$$S(x) = \frac{1}{1 + (x/x^*)^2}, \qquad x = \frac{c^2\|\nabla\nabla\Phi_N\|}{a_0^2}.$$

The effective interpolating function that multiplies the baryonic acceleration does not see C,

$$\nu_{\rm eff} = 1 + S(x)\big(\nu_{\rm std}(y) - 1\big). \tag{2}$$

C appears in the shell stress,

$$W_{ij} = -\frac{c^2}{2}\,S'(x)\,\big(P(y) - C\big)\,\frac{H_{ij}}{H_n}, \tag{3}$$

where $P(y) = \int_0^y 2t\,(\nu_{\rm std} - 1)\,dt$. For a spherical shell the outward piece is

$$g_W = \frac{2}{\sqrt{6}}\,r^{-3}\,\frac{d}{dr}(r^3\tilde{w}).$$

The frozen constants are $a_0 = 1.1607 \times 10^{-10}\,\mathrm{m\,s^{-2}}$ and $x^* = 7.035884281332817 \times 10^{11}$. Neither is fitted in the tests below. The algebra and the numerical match are derived [D] (`SCREENED_C_SOLUTIONS.json`, Res-Nova `ba3f6f6`; window check, Res-Nova `a223add`, Hood `67cc6c9`).

In plain words: S turns the extra gravity off where the tide is strong and leaves it on where the tide is weak. C slides the shell of transition without changing the rotation-curve law.

2 Why one field is not enough

Call $\eta = |\nabla\chi|/a_0$. A single scalar χ has one value of η at one point. Cassini, read as a Galactic external field at the Sun, needs η between 1.637 and 2.068. The Milky Way, read as a galaxy sitting in the field of M31, needs $\eta = 3.0 \times 10^{-3}$ at that same point. Those differ by a factor of about 550 to 690. No single-valued χ has both gradients. This is a no-solution statement, not a fit. The quantity that does take both values is the tide: $x \approx 1.8 \times 10^{12}$ at the Sun's MOND radius and $x \approx 3.9 \times 10^{7}$ in the disk at the solar circle, so $S \approx 0.133$ and $S \approx 1$ respectively.

3 The Oort window fixes C

For a uniform external field the binding offset of a comet that falls from rest is linear in C,

$$\frac{1}{a}(C) = \frac{1}{a}(0)\,\frac{P(y_e) - C}{P(y_e)}. \tag{4}$$

At $g_e = 1.90 \times 10^{-10}\,\mathrm{m\,s^{-2}}$, $P = 0.750345$ and $(1/a)(0) = 1.385818 \times 10^{-4}\,\mathrm{au^{-1}}$. The edge sits at 7,216 au and the radial gravity points outward between 8.77 and 11.16 kAU. The observed Oort peak lies at 10^4–10^5 au. Equation (4) puts that window at

$$C \in [0.208899,\ 0.696200],$$

with the centre $C = 0.579124707982253$ at 31,623 au. At $g_e = 2.32 \times 10^{-10}\,\mathrm{m\,s^{-2}}$ the window is $[0.384925,\ 0.872226]$ and the centre is 0.755150. These are $[D]$, and they match the stored solver to an error of 0 (Res-Nova `a223add`).

The working value used in every test below is the centre at 1.90×10^{-10}. This centre is the W-only osculating orbit. A full integration of the comet map with the divergence piece included, at this nonzero C, has not been run. That integration is open $[O]$.

4 Five tests at fixed C

Cassini. The action-form quadrupole at the same x^*, in units of $10^{-27}\,\mathrm{s^{-2}}$, is 5.200000 at $g_e = 1.9 \times 10^{-10}$ and 3.991075 at 2.4×10^{-10}, both at $C = 0$ and at the working C. Both sit inside the two-sided window $[-2.0,\ 5.2]$. The W parts differ by about 2×10^{-15} relative. A constant C integrates to $-2.3 \times 10^{-42}\,\mathrm{s^{-2}}$ against a $P(y)$ piece of 4.44×10^{-27}. The quadrupole is carried by the radial variation of P between 2.8×10^{3} and 1.25×10^{4} au. $[D]$. Res-Nova `a8e149d`, Hood `1159cf55`.

Gaia wide binaries. On one $0.8\,M_\odot$ binary the shell shrinks toward Newton: the dip goes from 0.901 to 0.986 at 8.0 kAU, and the peak goes from 1.343 to 1.134 at 17.5 kAU (Res-Nova `464ff2b`, Hood `1e96fc8e`). The catalogue test uses the full mass grid at $g_e = 1.9 \times 10^{-10}$ only. The clean cut, 14,542 pairs, gives raw $\chi^2 = 4.360$ and calibrated 2.995, against Newton 9.508 and an injection maximum of 20.431. The loose cut, 20,156 pairs, gives raw 39.605 and calibrated 18.171, under the bar of 18.5. The pre-registered exclusion does not fire. $[D]$. Res-Nova `1efb8fc`, Hood `d04f4f74`. The 2.4×10^{-10} column was not rerun.

SPARC. One hundred seventy-one galaxies, disk thickness 0.1 kpc, the same mass-to-light and distance fit as the earlier tidal run. The screen total is 18123.519. Newton in that fitter, with the extra gravity switched off, is 293096.718. The difference is -274973, so the screen is not worse than Newton. The unscreened $\mu_{\rm std}$ total is 18123.52096145091, and the screen sits 0.002 below it. No galaxy is an outlier under the pre-registered rule. Pre-registration Res-Nova `cb660b6`. Result $[D]$, Res-Nova `cc53735`, Hood `d2d980f6`. The shell term g_W does not move the aggregate. A disk solution of W, as opposed to the spherical estimate, is not in this number.

Four dwarfs. Crater II, Carina, Leo II, and Sculptor, at the measured dispersion minus its lower error, Milky Way speed 200 km s^{-1}. The decision mass-to-light ratios are 0.0599, 0.929, 2.659, and 1.335. All four are at or under 3, which is the pass line. The Newtonian control matches the published Newtonian ratios to a relative difference of 3.2×10^{-7}. Pre-registration Res-Nova `0c52fe8`. Result $[D]$, Res-Nova `b3a20b8`, Hood `a8016d1c`.

Leo II is not brought down by W. At the decision mass, g_W is $-8.0 \times 10^{-20}\,\mathrm{m\,s^{-2}}$. The ratio 2.659 is what the Wolf formula gives when the acceleration is $\nu_{\rm eff} g_N$ and S is about 1. An earlier isolated-branch formula, used for a different interpolating function, gave 6.558 for the same galaxy. Those are different estimators. The pass belongs to the Wolf formula written in the pre-registration.

5 What this paper does not claim

$C = 0$ fails the Oort test. That failure is a separate paper. An intermediate wide-binary cut, defined halfway between the clean and loose masks, is not a nested sample: 14,542, 18,053, and 20,156 pairs, with 35 clean pairs missing from the middle sample. Under the rule frozen for that cut, no exclusion is claimed and no model χ^2 was computed. $[D]$ for the counts, Res-Nova `d3ff415`, Hood `da07d02b`. The branch choice among relativistic completions is open $[O]$. x^* is the Cassini value, not a derived constant.

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